Distribution of electrical energy

1. You have two choices for transmitting 1.0×10^6 W of power through a long transmission line with 2.0 Ω of resistance:

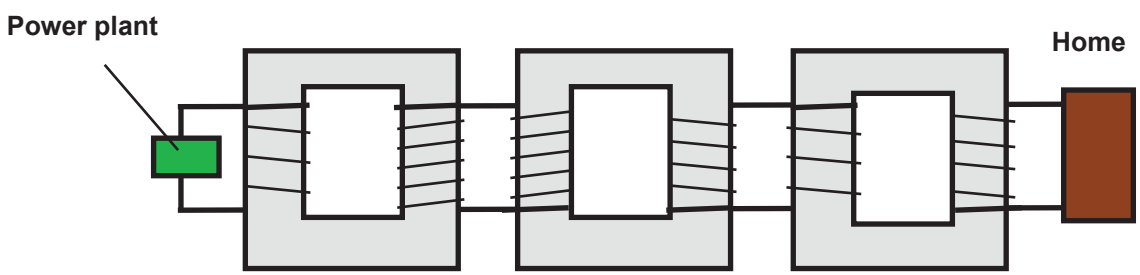
Choice 1: V = 10 000 V Choice 2: V = 800 000 V

Which one is the better choice? Explain your reasoning.

Solution:

Choice 2 will be more efficient because the higher the voltage, the less energy is lost during transmission.

2. The following diagram represents a simplified version of the power distribution grid.



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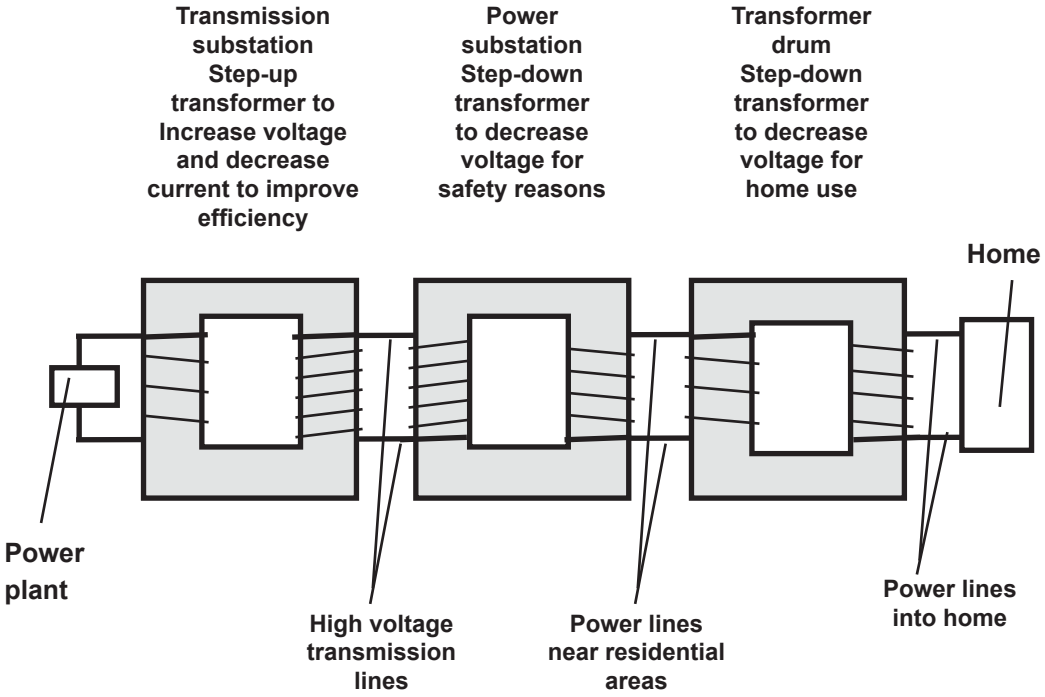
Draw your own copy of the diagram on a piece of paper. Label what each part of the diagram represents and explain why it is used.

Solution



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Explain, in your own words, why each of the following is essential to an efficient power distribution grid:

a. Transformers

Solution:

Transformers are needed to increase the voltage and lower the current from the generator in order to make the transmission of power more efficient. They are also need to lower the voltage and increase the current as the electricity is delivered to homes.

b. An AC current

Solution:

AC current creates a changing magnetic field, so transformers can be used. If DC current were used, the magnetic field would not change and transformers would not be able to produce a current in the secondary coil.

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